



BHARATIYA VIDYA BHAVAN
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI – 400 058.
Department of Computer Engineering

Mini Project (Semester V)

ACADEMIC YEAR: 2026-27

Date: 06/05/2026

Project Title: GraphMind: A Knowledge Notebook with Graph-Based Memory for Intelligent Document Question Answering

Abstract:

- **Problem:** Professionals (lawyers, doctors, researchers) waste hours searching through PDFs, scanned reports, and documents. Tools like Google's NotebookLM use vector-based search, which finds similar text but fails at connecting information across documents (e.g., "How does finding in Report A contradict Report B?").
- **Gap:** Vector-based RAG (Retrieval-Augmented Generation) treats documents as isolated text chunks. It cannot perform multi-hop reasoning, understand relationships between entities, or summarize themes across large collections.
- **Proposed Solution:** We build GraphMind — a document Q&A system that uses a knowledge graph as its memory instead of a flat vector database. When documents are uploaded, the system extracts entities (people, concepts, dates) and their relationships, building a graph that grows with each document. For answering queries, an intelligent router decides whether to use semantic search (vector), relational search (graph traversal), or both.
- **Key Features:**
 - (1) Automatic knowledge graph construction from uploaded documents using LLMs.
 - (2) Hybrid retrieval — vector search for simple queries, graph traversal for multi-hop queries.
 - (3) Multimodal support — handles text PDFs and scanned images.
 - (4) Source citations with every answer.
 - (5) Runs locally — no data leaves the user's machine.
- **Expected Outcome:** A working web application where users upload documents, ask natural language questions, and receive accurate, cited answers. We benchmark against standard vector-RAG and target 25–40% improvement on multi-hop queries.

References:

1. Pan, S., Luo, L., et al. "Unifying Large Language Models and Knowledge Graphs: A Roadmap." IEEE Trans. on Knowledge and Data Engineering (TKDE), Vol. 36, No. 7, 2024.
2. Jia, N., et al. "Knowledge Graph Enhanced RAG for Failure Mode and Effects Analysis." IEEE Trans. on Knowledge and Data Engineering (TKDE), 2024.
3. Wu, J., et al. "Graph Neural Networks for Information Retrieval: A Survey." IEEE Trans. on Neural Networks and Learning Systems (TNNLS), 2023.
4. Zhao, W., et al. "A Comprehensive Survey on Document Understanding with Deep Learning." IEEE Trans. on Pattern Analysis and Machine Intelligence (TPAMI), 2023.
5. Gao, Y., et al. "A Survey on Retrieval-Augmented Generation for Large Language Models." IEEE Access, 2024.
6. Edge, D., Trinh, H., et al. "From Local to Global: A Graph RAG Approach to Query-Focused Summarization." Microsoft Research, arXiv:2404.16130, 2024.
7. Zhu, Y., et al. "A Survey on Multimodal Knowledge Graphs: Construction, Completion, and Applications." IEEE/CAA Journal of Automatica Sinica, Vol. 11, No. 2, 2024.
8. Li, J., et al. "Retrieval-Augmented Generation for AI-Generated Content: A Survey." IEEE Intl. Conf. on Big Data, 2024.
9. Chen, W., et al. "Entity Extraction and Knowledge Graph Construction from Unstructured Text Using LLMs." IEEE Access, 2024.
10. Zhang, R., et al. "Question Answering over Knowledge Graphs with LLMs: A Survey." IEEE DSAA Conference, 2024.
11. Fan, T., et al. "MiniRAG: Towards Extremely Simple Retrieval-Augmented Generation." IEEE/ACM Proceedings, arXiv:2501.06713, 2025.
12. Kumar, S., et al. "Analyzing Embedding Models for Vector Databases." IEEE ICTBIG Conference, 2023.

Whether Interdisciplinary (Yes/No): No

If Yes, Name and sign of the Expert Guide from another Department: NA

Name & Signature: **Name of member 1: Bhumika A. Tarmale 2024300259**

Name of member 2: Sahil A. Valvi 2024300246

Name of member 3: Pranav B. Solunke 2024300246

Name and Signature of Co-Guide (if any): NA

Name and Signature of Guide: Prof. Yogita Patil